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Vectorized Interdisciplinary Overlap between STEM Domains

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Multidisciplinary Motivation



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- ▶ Students exercise *agency* in course selection beyond their declared major
- ▶ Cross-departmental enrollment signals overlap between programs
- ▶ NSF identifies convergence research as a top 10 BIG Idea: grand challenges “will not be solved by one discipline alone”
- ▶ Interdisciplinary programs can broaden participation for students with marginalized identities

What is MIDFIELD?



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- ▶ Multiple-Institution Database for Investigating Engineering Longitudinal Development
- ▶ Academic records of 98,000+ students from 18 US universities
- ▶ Data from 2000 onward to reflect contemporary curricula
- ▶ Contains: degree records, course enrollment, grades, demographics



- ▶ Shift unit of analysis from *student pathways* to *program relationships*
- ▶ Student enrollment as expression of agency and interdisciplinary interest
- ▶ Complementary similarity metrics applied to enrollment vectors
- ▶ Define three interpretable relationship types:
 - ▶ **Nested**: one program's coursework is largely contained within another
 - ▶ **Parallel**: shared base, divergent specialization
 - ▶ **Convergent**: deep overlap in both general and specialized content
- ▶ Extend curricular analytics *across* disciplinary boundaries



RQ: What types of curricular relationships have emerged between undergraduate STEM programs as evidenced by student enrollment patterns?

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From Student Records to Program Vectors



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1. Identify all courses taken by students completing each CIP code
2. Count course frequencies across all graduates of that major
3. Result: a high-dimensional *enrollment vector* per program
4. Apply similarity metrics to compare vectors pairwise

Why Multiple Metrics?



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- ▶ A single metric cannot distinguish:
 - ▶ Broad gen-ed overlap vs. specialized content overlap
 - ▶ Subset containment vs. symmetric overlap
 - ▶ Expected vs. statistically surprising co-enrollment

Jaccard Index (Curricular Width)

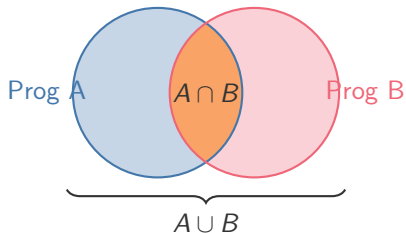


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$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

How much total curricular territory do two programs share?

- ▶ Treats every course equally
- ▶ Diluted by rare electives in large programs

TF-IDF Cosine (Specialized Identity)



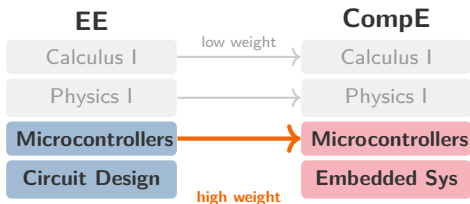
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$$\cos(\vec{a}, \vec{b}) = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

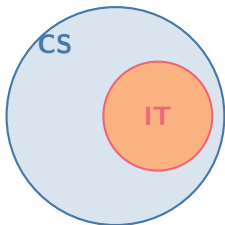


where weights use TF-IDF: upweights courses unique to few programs; downweights ubiquitous ones

Do two programs share identity-defining coursework?

- ▶ Resistant to program size differences
- ▶ Isolates specialized content from gen-ed noise

Overlap Coefficient (Subset Detection)



Overlap $\approx 80\%$
Jaccard $\approx 8.5\%$

$$\text{Overlap}(A, B) = \frac{|A \cap B|}{\min(|A|, |B|)}$$

Is one program's curriculum largely a subset of another's?

- ▶ Divides by the *smaller* set
- ▶ High Overlap + Low Jaccard = **nested**

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PMI (Signature Bonds)



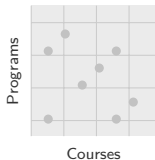
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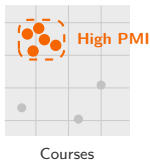
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Expected



Observed



$$\text{PMI}(m, c) = \log \frac{p(m, c)}{p(m) p(c)}$$

Do two programs share statistically surprising courses?

- ▶ Compares observed co-occurrence to expected
- ▶ High PMI = courses that “belong together”

Kulczynski (Mutual Resemblance)

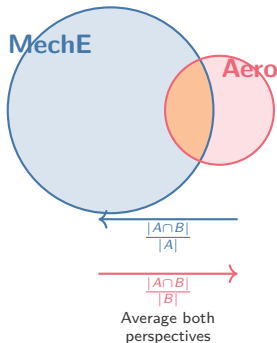


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$$\text{Kulczynski}(A, B) = \frac{1}{2} \left(\frac{|A \cap B|}{|A|} + \frac{|A \cap B|}{|B|} \right)$$

How well would a student from one program fit in the other?

- ▶ Averages overlap from each perspective
- ▶ Robust for large vs. small programs



Table: Metric signatures for three curricular relationship types

Type	Jaccard	TF-IDF	Overlap	PMI	Example
Nested	Low	Low	High	Var.	$IT \subset CS$
Parallel	Med	Low	Med	Low	$EE \parallel \text{Civil Eng}$
Convergent	High	High	High	High	$\text{Aero} \approx \text{Ind Eng}$

- ▶ The *pattern* across metrics defines the relationship, not any single score
- ▶ Relative values within a domain matter more than absolute percentages

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STEM Overview: Mathematics as a Bridge



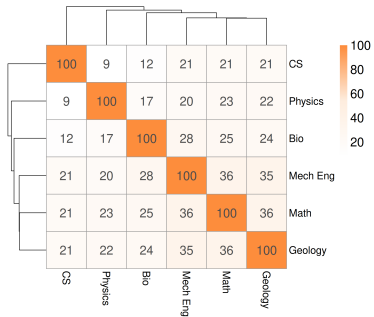
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**STEM Branches : Jaccard
(Curricular Width)**



- ▶ Math shares coursework with Mech Eng (~36%), Physics (~23%), CS (~21%)
- ▶ *Generalist connector* across STEM
- ▶ Validates the approach

STEM Overview: Nested Patterns Emerge

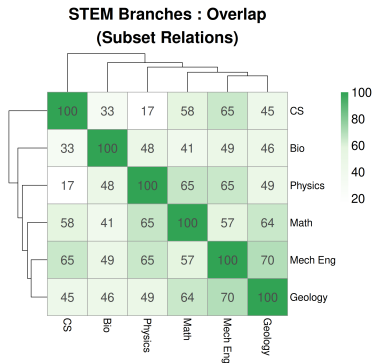


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- ▶ Physics–Mech Eng: ~65% Overlap vs. ~20% Jaccard
- ▶ High Overlap + Low Jaccard = **nested** relationship
- ▶ Physics coursework is largely contained within Mech Eng
- ▶ A single metric would miss this asymmetry

Engineering: The Parallel Pattern

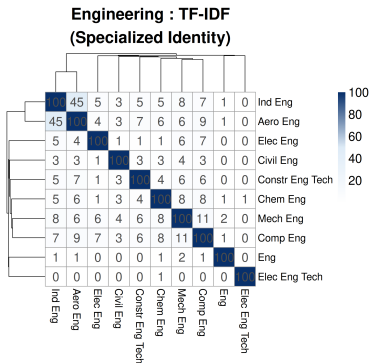


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- ▶ Median TF-IDF across engineering: ~3%
- ▶ Engineering programs share a large common core...
- ▶ ... but specialized content is very different
- ▶ This is the **parallel** pattern: shared base, divergent identity
- ▶ Exception: EE–CompE share ~33% Jaccard + high TF-IDF (merged departments)

Engineering: Aerospace–Industrial Convergence

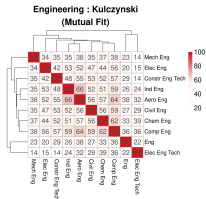
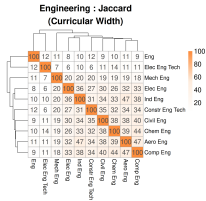


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► **Flagship result:** Aero–Industrial scores highest on every metric:

- TF-IDF ~45%, Jaccard ~47%
- Kulczynski ~66%, Overlap ~79%

► Textbook **convergent** relationship

► No one groups these administratively

Sciences: Cross-CIP Clustering

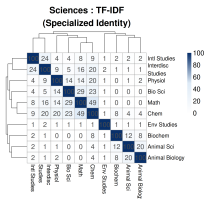
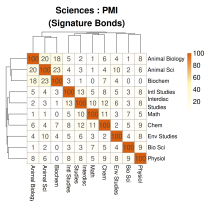


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- ▶ Animal Sci (CIP 01) clusters with Biochemistry & Animal Bio (CIP 26)
- ▶ Strongest PMI: Animal Sci–Biochem (~23%)
- ▶ Crosses CIP family boundaries
- ▶ Math–Chem TF-IDF (~49%): strongest specialized pair

Computing: The Nested Hierarchy

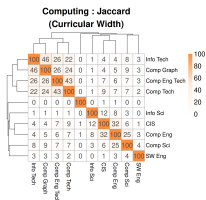
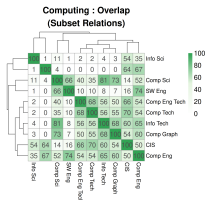


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► IT–CS: **cleanest nested example** in the dataset

- Overlap: ~80%
- Jaccard: ~8.5%
- Largest divergence between these metrics anywhere

► CS–CompE: mixed nested/convergent (Overlap ~52%, TF-IDF ~29%)

► Computing is organized as *nested hierarchies* branching from CS



- ▶ All metrics, domains, and program pairs
- ▶ Heatmaps with dendrograms
- ▶ Computing, engineering, sciences, cross-domain

<https://ransomts.github.io/ASEE-2026/>

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Key Takeaways



- ▶ STEM domains are interdisciplinary in *structurally different ways*
 - ▶ **Engineering**: parallel (shared core, divergent specialization)
 - ▶ **Computing**: nested hierarchies from CS
 - ▶ **Sciences**: cross-boundary clusters
- ▶ One-size-fits-all policy won't work

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Practical Implications



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- ▶ **Nested** ($IT \subset CS$): guide students toward parent field's electives
- ▶ **Parallel** (engineering): coordinate shared prerequisites
- ▶ **Convergent** (Aero $\approx$ Ind Eng): cross-listed courses, certificates, dual degrees

Future Work



- ▶ Disaggregate required vs. elective coursework
- ▶ Time series: are programs converging or diverging?
- ▶ Institutional disaggregation across universities
- ▶ Validate against known interdisciplinary programs

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